

Home Work 03: General Linear Fractional Quotient Derivative

Course: Mathematics-I (Differential Calculus)

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প্রদত্ত সমীকরণ: $y = \frac{5x + 3}{2x - 1}$

নির্ণেয়: $\frac{dy}{dx}$

ENGLISH NOTE — GENERAL LINEAR RATIONAL FORMULATION

The problem presents a linear rational transformation of the form $y = \frac{ax+b}{cx+d}$:

$$y = \frac{u(x)}{v(x)}, \quad \text{where } u(x) = 5x + 3, \quad v(x) = 2x - 1$$

Applying the fundamental Quotient Rule:

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}, \quad \text{with } \frac{d}{dx}(5x + 3) = 5, \quad \frac{d}{dx}(2x - 1) = 2$$

১. গাণিতিক কাঠামো ও প্রয়োজনীয় সূত্রসমূহ

উপাদান (Component)	ফাংশন রূপ	প্রমিত অন্তরজ সূত্র (Standard Derivative)
লব (u)	$5x + 3$	$\frac{d}{dx}(5x + 3) = 5$
হর (v)	$2x - 1$	$\frac{d}{dx}(2x - 1) = 2$
হরের বর্গ (v^2)	$(2x - 1)^2$	$(2x - 1)^2$ (উৎপাদক আকারেই রাখা সমীচীন)

ENGLISH NOTE — CRITICAL AMIE EXAM TRAPS & PITFALLS

1. Sign Distribution Across Expansion: Expand $-(5x + 3)(2) = -(10x + 6) = -10x - 6$. A classic student error is writing $-10x + 6$, which erroneously produces $+1$ instead of -11 .

2. Coefficient Preservation: Ensure linear coefficients 5 and 2 are correctly multiplied with corresponding binomial terms.

২. ধাপে ধাপে পূর্ণাঙ্গ সমাধান (Step-by-Step Solution)

ধাপ ১: উভয় পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করে:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{5x + 3}{2x - 1} \right)$$

ধাপ ২: ভাগফলের সূত্র (Quotient Rule) অনুসারে বিস্তার:

$$\frac{dy}{dx} = \frac{(2x - 1) \frac{d}{dx}(5x + 3) - (5x + 3) \frac{d}{dx}(2x - 1)}{(2x - 1)^2}$$

ধাপ ৩: প্রমিত অন্তরজের মান বসিয়ে:

$$\frac{dy}{dx} = \frac{(2x - 1) \cdot 5 - (5x + 3) \cdot 2}{(2x - 1)^2}$$

ধাপ ৪: লবের বন্ধনী বিস্তার ও পদসমূহ সরলীকরণ:

$$\frac{dy}{dx} = \frac{10x - 5 - 10x - 6}{(2x - 1)^2} = \frac{-11}{(2x - 1)^2}$$

চূড়ান্ত উত্তর (FINAL STANDARD ANSWER)

$$\frac{dy}{dx} = -\frac{11}{(2x - 1)^2}$$

 ENGLISH NOTE — DOMAIN, MONOTONICITY & SINGULARITY

- Domain of Function & Derivative:** Non-zero denominator requires $2x - 1 \neq 0 \implies x \neq \frac{1}{2}$. Thus, $\text{Dom}(y) = \text{Dom}\left(\frac{dy}{dx}\right) = \mathbb{R} \setminus \left\{\frac{1}{2}\right\}$.
- Strict Decreasing Monotonicity:** Because $(2x - 1)^2 > 0$ for all $x \neq \frac{1}{2}$ and numerator is negative ($-11 < 0$), $\frac{dy}{dx} < 0$ everywhere on its domain. The function is strictly decreasing with no local extrema.

 ENGLISH NOTE — DETERMINANT SHORTCUT FOR BILINEAR FUNCTIONS

For any linear fractional function $y = \frac{ax+b}{cx+d}$, the derivative equals:

$$\frac{dy}{dx} = \frac{ad - bc}{(cx + d)^2}$$

Here $a = 5, b = 3, c = 2, d = -1$:

$$ad - bc = (5)(-1) - (3)(2) = -5 - 6 = -11 \implies \frac{dy}{dx} = -\frac{11}{(2x - 1)^2}$$

This 5-second mental rule enables immediate verification!

 AMIE EXAM MARKS DISTRIBUTION STRATEGY

Exam Evaluation Step	Required Mathematical Rigor	Marks
Quotient Rule Formulation	Correct expansion of $\frac{vu' - uv'}{v^2}$	1.5
Linear Differentiation	Accurate coefficients $\frac{d}{dx}(5x + 3) = 5, \frac{d}{dx}(2x - 1) = 2$	1.5
Algebraic Simplification	Sign expansion $10x - 5 - 10x - 6 = -11$	1.0
Final Standard Format	Writing $-\frac{11}{(2x-1)^2}$	1.0
Total AMIE Section-A Marks		5.0 / 5.0